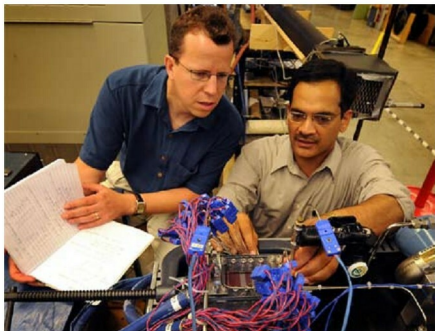


SOLIDSTATE REFRIGERATION WITH NO COMPRESSORS

In the future, it is possible that the cooling power of a noisy, bulky household refrigerator will be available on a small solidstate device with no compressor and heating coils



Researchers developing a miniature refrigeration system small enough to fit inside laptops and personal computers—a cooling technology that would boost performance while shrinking the size of refrigerators, too (Courtesy: Purdue News Service photo/David Umberger)

■ DR S.S. VERMA

A symbol of modern living, refrigeration has changed our lives completely. It has become an integral part of our lifestyle and revolutionised our homes and commercial places with its applications in air-conditioning and refrigerating foodstuffs in homes, restaurants and large storage warehouses. Its use in our kitchens for storing fruits and vegetables has allowed us to add fresh salads to our diets year round, and to store fish and meats safely for long periods.

However, for the last few decades, there has not been any significant in-

crease in the efficiency of the existing compressor-based refrigeration—mechanical refrigeration system—which has reached the maximum level of innovation. Moreover, with the increasing awareness for environmental degradation, the production, use and disposal of chlorofluorocarbons (CFCs) and hydro chlorofluorocarbons (HCFCs) as refrigerants in mechanical refrigeration system has become an issue of great concern. Therefore efforts are being made to develop eco-friendly alternative refrigeration systems. Researchers who are investigating other methods of refrigeration are expecting that conventional refrigerators and other

cooling devices may one day lose their compressors, coils of piping, harmful refrigerants and become solidstate.

Refrigeration is the process of removing heat from an enclosed space, or from a substance, and rejecting it elsewhere for the primary purpose of lowering the temperature of the enclosed space or substance and then maintaining that lower temperature. Cooling is the natural or artificial process by which heat is dissipated. Cold is the absence of heat, hence in order to decrease temperature, one removes heat, rather than adding cold. In order to satisfy the Second Law of Thermodynamics, some form of work must be performed. This work is traditionally done mechanically but can also be done by other means.

In commerce and manufacturing, there are many uses for refrigeration. It is used for liquifying gases like oxygen, nitrogen, propane and methane. In compressed air purification, it is used for condensing water vapour from compressed air to reduce its moisture content. In oil refineries, chemical plants and petrochemical plants, refrigeration is used for maintaining certain processes at their required low temperatures. Metal workers use refrigeration to temper steel and cutlery. In transporting temperature-sensitive foodstuffs and other materials by trucks, trains, airplanes and sea-going vessels, refrigeration is a necessity.

Conventional refrigeration: Causing concern

The basic idea behind a refrigerator is to use the evaporation of a liquid